WHITE DWARFS · QUANTUM MECHANICS, GRAVITY AND RELATIVITY

The Chandrasekhar limit

$\simeq 1.44\,M_\odot$

A mass limit appears when two energies grow at the same rate under compression.

Model contract. A white dwarf is a compact stellar remnant. Its nuclei carry almost all the mass, and its electrons provide the pressure. Take a cold, spherical, non-rotating star of mass M and radius R, with Newtonian gravity. Write $M = N\mu_e m_\mathrm{p}$, where N is the electron count, μ_e is the number of nucleons per electron, and m_p is the proton mass, used as the mass per nucleon. Fully ionised carbon and oxygen both have $\mu_e = 2$.

1 / QUANTUM SUPPORT

Squeezing raises electron momentum

Even at zero temperature, electrons occupy distinct states by the Pauli principle, filling them up to the **Fermi momentum** p_F. At number density $n_e = N/V$, their mean spacing is $\ell \sim n_e^{-1/3}$. Use $\Delta x\,\Delta p \geq \hbar/2$ with $\Delta x \sim \ell$ to estimate the momentum scale:

$$p_F \sim \frac{\hbar}{\ell} \sim \frac{\hbar N^{1/3}}{R}, \qquad V = \frac{4\pi R^3}{3}.$$

Smaller R means larger p_F, so compression costs kinetic energy.

2 / GRAVITATIONAL BINDING

Squeezing lowers gravitational energy

Bring a shell of mass $\mathrm{d}m$ from infinity to radius r. The mass $m(r)$ already inside it gives $\mathrm{d}U_g = -Gm(r)\,\mathrm{d}m/r$. Adding the shells gives

$$U_g = -B\,\frac{GM^2}{R},$$

where the number B depends on how density varies with radius. For a sphere of uniform density, $B = 3/5$.

3 / HOW MUCH ENERGY DOES COMPRESSION COST?

Special relativity changes the power of R

An electron of momentum p has kinetic energy $\epsilon = \sqrt{(m_\mathrm{e}c^2)^2 + (pc)^2} - m_\mathrm{e}c^2$. Estimate the total by $U_e \sim N\epsilon(p_F)$.

Electron regime	One electron	Total electron energy	Halve R at fixed M
$p_F \ll m_\mathrm{e}c$ (slow)	$\epsilon \simeq \dfrac{p^2}{2m_\mathrm{e}}$	$U_e \sim \dfrac{\hbar^2 N^{5/3}}{m_\mathrm{e}R^2}$	U_e quadruples
$p_F \gg m_\mathrm{e}c$ (ultrarelativistic)	$\epsilon \simeq pc$	$U_e \sim \dfrac{\hbar c\, N^{4/3}}{R}$	U_e doubles

The binding energy $|U_g|$ doubles in both cases. If electrons stayed slow, the $+1/R^2$ term would eventually win over $-1/R$, giving a finite energy minimum for any mass. In the ultrarelativistic limit, both terms scale as $1/R$.

4 / BALANCE THE ENERGIES IN THE RELATIVISTIC LIMIT

The radius cancels, leaving a mass threshold

For a fixed density shape, A and B are dimensionless factors. Substitute $N = M/(\mu_e m_\mathrm{p})$ into the total energy:

$$U_e + U_g \simeq \frac{1}{R}\left[A\hbar c\left(\frac{M}{\mu_e m_\mathrm{p}}\right)^{4/3} - BGM^2\right].$$

At the threshold the bracket is zero. Divide by $M^{4/3}$, then raise to $3/2$:

$$M^{2/3} = \frac{A\hbar c}{BG(\mu_e m_\mathrm{p})^{4/3}} \quad\Longrightarrow\quad M_\mathrm{lim} = CM_0, \qquad C = (A/B)^{3/2}, \qquad M_0 = \frac{(\hbar c/G)^{3/2}}{(\mu_e m_\mathrm{p})^2}.$$

As M rises, the gravitational term grows as M^2, faster than $M^{4/3}$. Above the threshold the bracket is negative: shrinking lowers the energy, so electron pressure cannot maintain a cold white dwarf.

5 / PUT BACK THE NUMERICAL FACTORS

Why the bare scale is $0.46\,M_\odot$ and the limit is about $1.44\,M_\odot$

Model	C	$M/M_\odot$
Bare mass scale M_0	1	0.46
Uniform sphere	3.72	1.72
Hydrostatic profile	3.10	1.44

Values for $\mu_e = 2$; $M_\odot$ is the Sun's mass.

Uniform sphere. State counting gives $p_F = \hbar(3\pi^2 n_e)^{1/3}$ and the mean ultrarelativistic energy is $3p_F c/4$. This gives $A = \frac{3}{4}(9\pi/4)^{1/3}$.

Vary the density. At fixed N, $U_e \propto V^{-1/3}$ gives $P = -\partial U_e/\partial V \propto \rho^{4/3}$, where $\rho = M/V$. The index 3 Lane–Emden solution of hydrostatic balance gives $C = \sqrt{3\pi}\,\omega_3/2 \simeq 3.10$ with $\omega_3 \simeq 2.018$.

Composition check. If μ_e rises from 2 to 2.5, the limit falls to $1.44(2/2.5)^2 \simeq 0.92\,M_\odot$. Fewer electrons per kilogram give less support at the same mass and radius.

Based on *Assembling the Chandrasekhar Limit*. Original model: Chandrasekhar, *The Maximum Mass of Ideal White Dwarfs* (1931).

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